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Contest (1)

template.cpp44 lines

```
#include <bits/stdc++.h>
using namespace std;
// #define LOCAL_DEBUG
#ifdef LOCAL_DEBUG
#define dbg(...) cerr<<"DEBUG:["<<#__VA_ARGS__<<"] = [" ,\
    _print(__VA_ARGS__),cerr<<"]\n"
template<class T, class=void>struct cntr : false_type {};
template<class T>struct cntr<T,
    void_t<decltype(declval<T>().begin()),
    decltype(declval<T>().end()),
    typename T::value_type> > : true_type {};
template<class T>using raw=remove_cv_t<remove_reference_t<T> >;
template<class T>enable_if_t<!cntr<T>::value>
    _print(const T &x) { cerr << x; }
template<class T>enable_if_t<cntr<T>::value>_print(const T &v){
    cerr << "["; bool f = false;
    for (const auto &x: v)
        f ? cerr << (cntr<raw<decltype(x)> >::value ? ",\n":", "):
            cerr, f = true, _print(x);
    cerr << "]" ;
}
template<class T, class... A>
void _print(const T &x, const A &... a) {
    _print(x);(cerr << " , ", _print(a)), ... ;}
#else
#define dbg(...) ;
#endif
#define rep(i, a, b) for(int i = a; i < (b); ++i)
#define all(x) begin(x), end(x)
#define sz(x) (int)(x).size()
using ll = long long;
using pii = pair<int, int>;
using vi = vector<int>;
void init() {}
void solve() {}
signed main() {
    init();
    cin.tie(nullptr), ios::sync_with_stdio(false);
    // cin.exceptions(cin.failbit);
    int T = 1;
    // cin >> T;
    while (T-->solve());
    return 0;
}
```

Techniques (A)

techniques.txt

159 lines

Recursion
 Divide and conquer
 Finding interesting points in $N \log N$
 Algorithm analysis
 Master theorem
 Amortized time complexity
 Greedy algorithm
 Scheduling
 Max contiguous subvector sum
 Invariants
 Huffman encoding
 Graph theory
 Dynamic graphs (extra book-keeping)
 Breadth first search
 Depth first search
 * Normal trees / DFS trees
 Dijkstra's algorithm
 MST: Prim's algorithm
 Bellman-Ford
 Konig's theorem and vertex cover
 Min-cost max flow
 Lovasz toggle
 Matrix tree theorem
 Maximal matching, general graphs
 Hopcroft-Karp
 Hall's marriage theorem
 Graphical sequences
 Floyd-Warshall
 Euler cycles
 Flow networks
 * Augmenting paths
 * Edmonds-Karp
 Bipartite matching
 Min. path cover
 Topological sorting
 Strongly connected components
 2-SAT
 Cut vertices, cut-edges and biconnected components
 Edge coloring
 * Trees
 Vertex coloring
 * Bipartite graphs ($\Rightarrow$ trees)
 * $3 \cdot n$ (special case of set cover)
 Diameter and centroid
 K'th shortest path
 Shortest cycle
 Dynamic programming
 Knapsack
 Coin change
 Longest common subsequence
 Longest increasing subsequence
 Number of paths in a dag
 Shortest path in a dag
 Dynprog over intervals
 Dynprog over subsets
 Dynprog over probabilities
 Dynprog over trees
 3^n set cover
 Divide and conquer
 Knuth optimization
 Convex hull optimizations
 RMQ (sparse table a.k.a 2^k -jumps)
 Bitonic cycle
 Log partitioning (loop over most restricted)
 Combinatorics

Computation of binomial coefficients
 Pigeon-hole principle
 Inclusion/exclusion
 Catalan number
 Pick's theorem
 Number theory
 Integer parts
 Divisibility
 Euclidean algorithm
 Modular arithmetic
 * Modular multiplication
 * Modular inverses
 * Modular exponentiation by squaring
 Chinese remainder theorem
 Fermat's little theorem
 Euler's theorem
 Phi function
 Frobenius number
 Quadratic reciprocity
 Pollard-Rho
 Miller-Rabin
 Hensel lifting
 Vieta root jumping
 Game theory
 Combinatorial games
 Game trees
 Mini-max
 Nim
 Games on graphs
 Games on graphs with loops
 Grundy numbers
 Bipartite games without repetition
 General games without repetition
 Alpha-beta pruning
 Probability theory
 Optimization
 Binary search
 Ternary search
 Unimodality and convex functions
 Binary search on derivative
 Numerical methods
 Numeric integration
 Newton's method
 Root-finding with binary/ternary search
 Golden section search
 Matrices
 Gaussian elimination
 Exponentiation by squaring
 Sorting
 Radix sort
 Geometry
 Coordinates and vectors
 * Cross product
 * Scalar product
 Convex hull
 Polygon cut
 Closest pair
 Coordinate-compression
 Quadrees
 KD-trees
 All segment-segment intersection
 Sweeping
 Discretization (convert to events and sweep)
 Angle sweeping
 Line sweeping
 Discrete second derivatives
 Strings
 Longest common substring
 Palindrome subsequences

Knuth-Morris-Pratt
 Tries
 Rolling polynomial hashes
 Suffix array
 Suffix tree
 Aho-Corasick
 Manacher's algorithm
 Letter position lists
 Combinatorial search
 Meet in the middle
 Brute-force with pruning
 Best-first (A^*)
 Bidirectional search
 Iterative deepening DFS / A^*
 Data structures
 LCA (2^k -jumps in trees in general)
 Pull/push-technique on trees
 Heavy-light decomposition
 Centroid decomposition
 Lazy propagation
 Self-balancing trees
 Convex hull trick (wcipeg.com/wiki/Convex_hull_trick)
 Monotone queues / monotone stacks / sliding queues
 Sliding queue using 2 stacks
 Persistent segment tree